Sonar for Astrophysics: The BATPING Project

(Bidirectional Acoustic Traveltime Pulser for INdex Gradients)

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I. Introduction

The Legacy Survey of Space and Time (LSST) program at the Vera C. Rubin Observatory, located in Chile's Cerro Pachón mountain range, will generate an unprecedented catalog of astronomical data to enable precise measurements of dark matter and dark energy in our universe. Understanding the nature of dark matter and the even more mysterious dark energy that together account for 95% of the energy/matter content of our universe is the forefront mission within astrophysics and cosmology today. LSST achieved First Light in July 2025, and is now generating 20 terabytes of digital imagery nightly. Harvard Physics Professor Christopher Stubbs is a longtime leader of LSST/Rubin and has served in major roles during its construction and commissioning.

I propose to conduct research aimed at addressing LSST image degradation from variable environmental conditions within the dome enclosure and near-atmosphere. I have been working with Professor Stubbs on related projects over the past year, namely: (1) computer vision-based detection and prediction of aircraft contrail streaks in summit sensor data; and (2) supplying the Auxiliary Telescope's photometric calibration system with a Raspberry Pi-based coordinate-finding module [1]. Continuing on this theme, Professor Stubbs and I have identified a synergistic project that we call BATPING (Bidirectional Acoustic Traveltime Pulser for Index Gradients). BATPING will allow us to characterize air temperature fluctuations within the telescope enclosure, so as to reduce associated image distortions.

II. Project Description

My work will follow two thrusts: (A) optimization and implementation of the BATPING hardware configuration, an array of piezo ultrasonic transmitters and acoustic thermometer receptors, and (B) development of a machine learning (ML) approach to optimize the acoustic signal processing. In the process of carrying out this research, I will also gain a significant understanding of image fidelity that will allow me to contribute to subsequent scientific analyses of LSST data. In the following, I describe the BATPING project, and my research plan for summer 2026.

A. Instrument Methodology

Temperature fluctuations within the telescope enclosure have the potential to perturb the optical path, decreasing image quality in an effect known as dome seeing. Because the index of refraction of air is temperature-dependent, turbulent pockets of warmer and colder air alike cause refraction of incoming wavefronts. This translates to a broadening of the telescope's point spread function (PSF), the size of the image in the focal plane that captures an ideal point source, like a star. The effect of dome seeing has been recently quantified by the Stubbs group through measurements of the Auxiliary Telescope's PSF full width at half-maximum (FWHM), which was found to be worse than the telescope's diffraction limit [2]. As shown in Figure 1, other members of the Stubbs group have identified this increase in FWHM

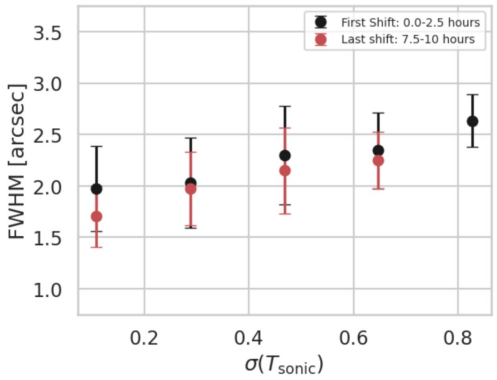


Fig. 1. Correcting for temperature variations to the level of 0.1° C will bring the telescope's point spread function (PSF) to within 10% of its diffraction-limited value. Just $\sim 0.5^\circ$ of temperature fluctuation broadens the PSF by $25-50\%$.

with fluctuations (characterized by the standard deviation $\sigma(T_{\text{sonic}})$) in the sonic temperature within the dome.

To measure and correct for $\sigma(T_{\text{sonic}})$, I propose an acoustics-based transmitter-sensor system. I envision using an array of 40kHz piezoelectric transmitters whose waveform is optimized based on the ML algorithm described below. A high-fidelity acoustic thermometer would then measure air temperature based on the difference in travel time taken along opposing paths by this ultrasonic pulse. Acoustic thermometers are attractive because they are insensitive to wind speed, solely measuring the contribution of temperature to the acoustic wave speed:

$$T_s \equiv \frac{c^2}{\gamma_d R_d}, \quad \begin{aligned} \gamma_d &= \text{dry air specific heat ratio,} \\ R_d &= \text{dry air gas constant} \end{aligned}$$

From the measured speed of sound, one can determine the standard deviation of path-averaged sonic temperature, a direct proxy for refractive index. An initial target is to detect $\sigma(T_{\text{sonic}})$ at the level of roughly 0.1° C [3].

B. Data Analysis and Machine Learning Methods

Processing the acoustic signal will be critical in ensuring accurate temperature readings, as the traveltime calculations rely on an easily identifiable narrow wavefront. Most notably, the impulse response $h(t)$ of the receiving transducer introduces to the transmitted waveform $s(t)$ a tail that is known as a ringdown:

$$y(t) = (s * h)(t - \tau) + n(t) \quad \text{for } n(t) \text{ noise,}$$

where $*$ denotes the convolution, τ is the traveltime, and $y(t)$ is the received waveform.

I will develop an ML-based algorithm to optimize the transmitted waveform $s_\theta(t)$ through parametrization θ, such that that the new received waveform $y(t;\theta)$ approaches the delta function and structured noise is eliminated:

$$y(t;\theta) = (s_\theta * h)(t - \tau) + n(t) \approx \delta(t - \tau) \qquad (1)$$

I have identified a data set containing ultrasound waveforms with and without ringdown to train the model [4]. During the training stage, the proposed ML model receives waveform segments and outputs a modified transmit waveform that is proposed to suppress ringdown and other noise. The loss function is defined as the proportion of the waveform energy contained within an expected time interval: more ringdown implies a spread of energy beyond the time of arrival.

In sum, the model learns the transducer impulse response and optimizes the sharpness of the output signal peak, enabling adaptive cancellation of the ringdown tail. The Python library **PyTorch**, with its associated add-on **torchaudio**, provides the relevant ML tools.

III. Project Impact

Assessments from the first cycle of LSST data indicate a need to further address image degradation. In particular, while optomechanical factors are being accounted for with high levels of precision, dome and mirror seeing remains uncharacterized [5]. A number of optics-based projects are being implemented to characterize the refractive index gradient along the optical path, commonly with a strobed light source oriented towards the telescope [6]. Complementing these efforts, BATPING offers the advantage that the speed of sound is much more sensitive to variations in temperature and hence refractive index. Additionally, BATPING's ultrasonic signals, even along the optical path, do not interfere with normal observations, nor do they rely on dome illumination.

Furthermore, the BATPING apparatus has the unique potential to extend analysis beyond the optical path. Computational Fluid Dynamics (CFD) simulations suggest that temperature measurement of the entire dome, particularly the telescope camera surface and enclosure walls, is a critical step in minimizing dome seeing effects and reaching the telescope's diffraction limit [7]. Establishing an array of BATPING's emitter-responders throughout the enclosure offers a promising system for performing this anticipated dome-wide tomography. The LSST observing program has only just started: image quality improvement efforts like BATPING will enhance its scientific reach over the long term.

IV. Research Schedule and Travel Plan

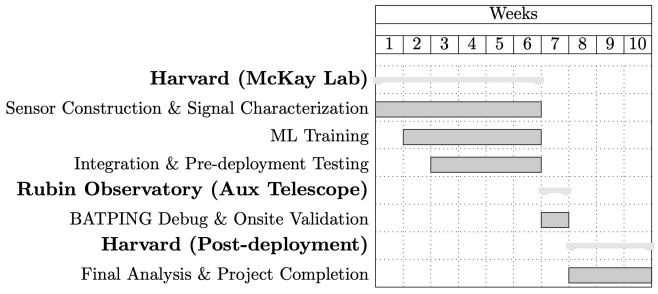


Fig. 2. Research Schedule

I plan to conduct the majority of my research in accordance with the schedule given in Figure 2. In addition to Prof. Stubbs, I will work with Prof. Paul Horowitz on electronics for BATPING, and with postdoctoral researcher Dr. Johnny Esteves on harnessing ML methods.

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